BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Paper 1: Networks and Security

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. How do you calculate raw text size in bits?

2. What can a subprogram's parameter receive?

A) Only integers
B) An array
C) Only strings
D) Only Boolean values

3. "readLine() writes a new line into the file." Correct this. Explain why.

4. A program saves a result to a text file. "writeLine(text) only shows text on screen." Correct this. How should the program save the result?

Today's objectives

Explain the core ideas of paper 1: networks and security using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Network choice

Distinguish LAN/WAN, client-server/peer-to-peer and suitable devices.

Internet services

DNS maps domain names to IP addresses; routers connect networks.

Recognise attacks

Distinguish phishing, malware and denial of service.

Prevent attacks

Combine least privilege, authentication, encryption, updates and backups.

Write one key term and its meaning in your own words:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Key ideas

Network choice	Distinguish LAN/WAN, client-server/peer-to-peer and suitable devices.
Internet services	DNS maps domain names to IP addresses; routers connect networks.
Recognise attacks	Distinguish phishing, malware and denial of service.
Prevent attacks	Combine least privilege, authentication, encryption, updates and backups.

I Do - One worked example

A school receives a phishing email directing staff to an unfamiliar login page. Recommend two defences.

1. Identify credential theft/social engineering.
2. Train staff to verify suspicious links through another channel.
3. Use MFA so a password alone is insufficient.
4. Explain both measures target different stages of compromise.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. A device has poor Wi-Fi but fast wired access. Explain two possible causes.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Correct the comparison so a warning appears on the third failed attempt.

```
failed = 3
if failed > 3 then
print("Review access")
endif
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. State one role of DNS. [2 marks]

Q3. Name the device that connects networks. [1 mark]

Q4. Explain why encryption alone does not prevent phishing. [3 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. Encryption stops all social engineering attacks. ☐ T ☐ F
2. Distinguish LAN/WAN, client-server/peer-to-peer and suitable devices. ☐ T ☐ F
3. DNS maps domain names to IP addresses; routers connect networks. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 Number of characters x bits per character, using the given encoding. ☐

Q2 B - An array ☐

Q3 It reads the next line and advances the current position. ☐

Q4 writeLine(text) writes a line through an open file handle. Open the file, write, then close it. Printing on screen does not save it. ☐

Code First - Model

```
failed = 3
if failed >= 3 then
print("Review access")
endif
```

Output: Review access

We Do - Check the model

1. Do not assume internet service itself must be slow. ☐

Exit Ticket - Check

1. FALSE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Resolves domain names to IP addresses. ☐

Q3 Router. ☐

Q4 Encryption protects data in transit but cannot stop a user willingly entering credentials into a fraudulent site. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: